Two Routes to Quantum Tunnelling

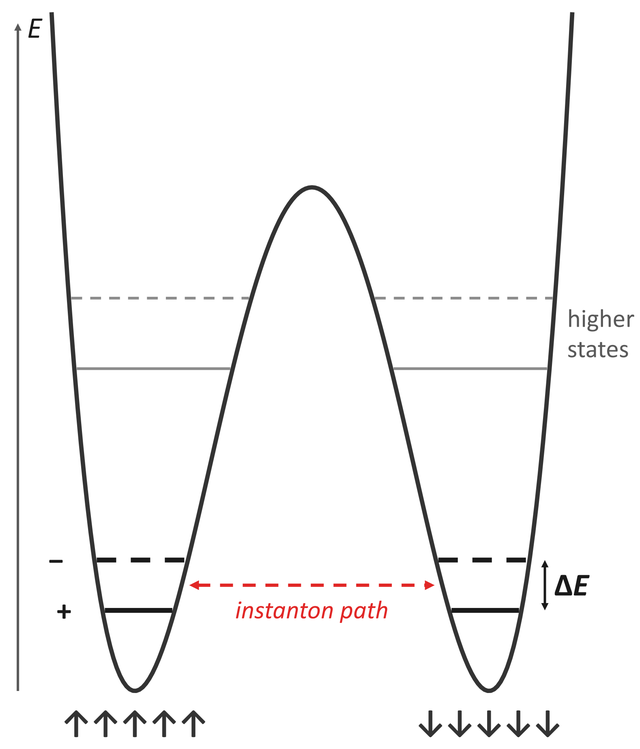
Numerical cost, instanton accuracy and the role of symmetry

Syed Hussain Haider, Qiongyuan Wu, James Millen
Department of Physics, King's College London, Strand, London WC2R 2LS UK
syedhussainhaider137@gmail.com

As tunnelling becomes weaker, do the useful ranges of numerical and semiclassical calculations meet?

ONE PHYSICAL QUANTITY, TWO ROUTES

Two classically equivalent configurations can be coupled by quantum tunnelling. The lowest symmetric and antisymmetric eigenstates form a doublet with splitting ΔE. This splitting becomes exponentially small as tunnelling is suppressed. Two methods are compared here.



Schematic double well with the lowest symmetric (+, solid) and antisymmetric (−, dashed) states. The splitting is enlarged.

Route 1: numerical imaginary-time evolution

$$|\psi(\tau)\rangle \propto \mathrm{e}^{-\hat{H}\tau/\hbar}\,|\psi(0)\rangle$$

Higher-energy components are suppressed by imaginary-time evolution. If both doublet states are present, their relative amplitude is reduced by a factor $\exp(-\Delta E\tau/\hbar)$. For a fixed reduction, a time of order $\boldsymbol{\hbar/\Delta E}$ is required. With uniform time steps, more work is needed for a smaller splitting.

The numerical error can be reduced by refining the calculation. The main difficulty is reaching convergence within the available time while keeping this error small.

Route 2: semiclassical instanton theory

$$\Delta E_{\mathrm{inst}} \simeq A(s)\,\mathrm{e}^{-s}, \qquad s = S_{\mathrm{inst}}/\hbar$$

Tunnelling through the forbidden region is described by the instanton action. A is the fluctuation prefactor. Leading-order theory is cheap to evaluate. It is accurate in the semiclassical regime, where tunnelling is weak, and becomes less accurate outside it [1,2].

Do the useful ranges meet?

Route 1 is limited mainly by calculation time. Leading Route 2 is limited by approximation error. At a chosen accuracy and time budget, their useful ranges may overlap or leave a coverage gap. This is different from the energy splitting ΔE.

TEST CASE: A COLLECTIVE ISING MODEL

The fully connected transverse-field Ising model is used:

$$\hat{H} = -2\Gamma\hat{S}_x - \frac{2}{L}\hat{S}_z^2$$

$$S = L/2, \qquad \Gamma = 0.5$$

L is the number of spins, and $S = L/2$ is their total spin. The interaction strength and $\hbar$ are set to one. With permutation symmetry, only $L + 1$ collective-spin states are needed. Smaller L gives stronger tunnelling, so imaginary-time evolution is faster. Larger L gives weaker tunnelling, so the instanton is more accurate.

First calculation: parity sectors not separated

Uniform-step Crank–Nicolson evolution is started in one magnetisation well. The splitting is obtained from magnetisation decay. The results remain accurate, but the runtime increases rapidly.

L	Numerical time (s)	Instanton error (%)
8	0.0418	16.72
16	1.93	6.19
20	13.01	4.68
24	86.76	3.77

Table 1. Numerical error < 0.0002% in every row. Uniform time steps are used without separating parity sectors. The recorded times depend on the machine.

Instanton error first falls below 1% at $L = 100$. At the $L = 24$ step rate, the original method would need ~1.3×10^{17} s (~4.2 billion years) to reach this size.

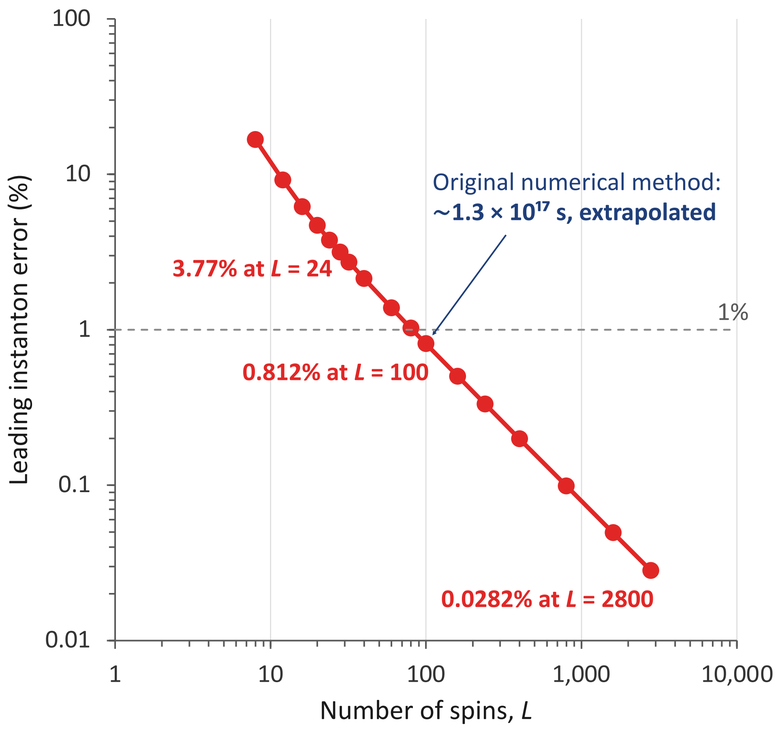


Figure 1. Leading instanton relative error at $\Gamma = 0.5$. A 1% accuracy target is shown. The first sampled point below it is $L = 100$.

With a 1% target and a short time budget, there is a range of L where neither method works. The original method becomes too slow soon after $L = 24$, but the instanton is not accurate enough until $L = 100$. This coverage gap is estimated from runs up to $L = 24$ at a fixed step rate. It applies to this method, not numerical methods in general.

THE TURNING POINT: USE PARITY

The Hamiltonian has exact spin-flip symmetry. The even and odd sectors can be evolved separately. Each doublet member is the lowest state in its sector. Its almost-degenerate partner is excluded. Convergence is controlled by higher states of the same parity.

The splitting ΔE is obtained from boundary amplitudes and an overlap using a discrete Herring/Wronskian identity [3]. This gives ΔE directly, instead of subtracting two almost identical energies, which a computer cannot do accurately. The Hamiltonian and splitting are unchanged.

L	Numerical time (s)	Instanton error (%)
160	0.0268	0.502
800	0.1956	0.0989
2800	1.6187	0.0282

Table 2. Uniform-step CN in separate parity sectors, with Wronskian readout. Times are recorded when 10^{-10} relative error is first reached against a reference. Instanton errors are leading order. The accuracies are not matched. These times do not give a certified stopping rule.

With parity separated, the numerical calculation remains cheap where leading instanton theory is accurate. At $L = 2800$, 10^{-10} relative error is reached in 1.62 seconds. The instanton error is 0.0282%. Different accuracies are reached, but the 1% target is met by both methods.

Illustrative criterion: 1% relative error, two-minute numerical budget.

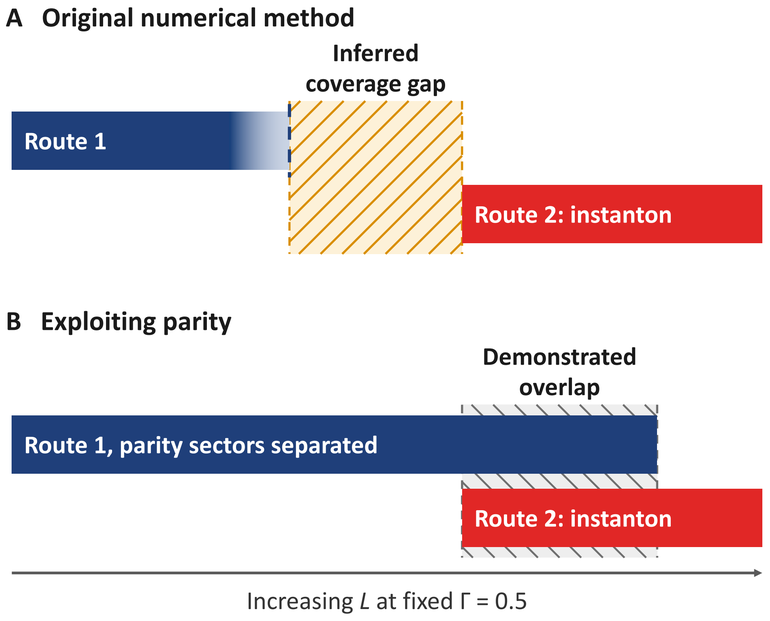


Figure 2. Schematic useful ranges of the two routes. The original boundaries are inferred from measured cost growth and extrapolation. They were not measured at matched tolerances. Overlap is shown by the parity results. The boundaries depend on the accuracy and time budget.

Conclusion

For this model, a strong overlap is obtained when parity is used with the Wronskian readout. Numerical results remain affordable where instanton theory is accurate enough. The useful ranges depend on the Hamiltonian, the numerical method and the accuracy required.

Future direction

Weakly broken symmetry could be studied next. Perturbative corrections to the symmetric solution could be tested to see whether the low cost is retained.

[1] S. V. Isakov *et al.*, Understanding quantum tunneling through quantum Monte Carlo simulations, *Phys. Rev. Lett.* **117**, 180402 (2016).
[2] A. Garg, *J. Math. Phys.* **39**, 5166 (1998), doi:10.1063/1.532563.
[3] C. Herring, *Rev. Mod. Phys.* **34**, 631 (1962).
[4] S. H. Haider, Two Routes to Quantum Tunnelling: poster code, GitHub (2026). github.com/syedhussainhaider137/tunnelling-splitting-poster